A short note about the seminar of numerical analysis for dynamical systems

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Χάος μὲν πρῶτον γένετο, Γαῖα δὲ εὐρύστερνος,
πάντων ἕδος ἀσφαλὲς ἀθανάτων, οἳ ἔχουσι κάρηνα
Ὄλυμπον χιονόεντα, Τάρταρος ἠερόεντα
τεθλαμένος, Ἔρος δὲ καλλίστατος ἐν θεοῖσι.

At first Chaos came to be, but next wide-bosomed Earth,
the ever-sure foundation of all the deathless ones who hold
the peaks of snowy Olympus, and dim Tartarus
in the depth of the wide-pathed Earth, and Eros (Love),
fairest among the deathless gods, who unnerves the limbs
and overcomes the mind and wise counsels of all gods and all men.

Hésiode, *Théogonie*, 116-120 (translated by H.G. Evelyn-White, 1914).

Abstract

This is a small introductive note related to the seminar of numerical analysis for differential equations and dynamical systems (Udelar, PEDECIBA, 2026).

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1 Introduction (more or less autobiographic)

I write this note as a kind of entry point to the wonderful universe of numerical analysis related to dynamical systems in a broad sense, with emphasis on ordinary differential equations with geometric properties and the calculation of bifurcations. These notes are motivated by a seminar on this topic I am organizing in 2026 at the Udelar (Montevideo, Uruguay) for undergraduate and graduate students (in the framework of the Pedeciba program for graduate and doctoral studies). They are also inspired by a lot of discussions I had, by some talks I attended, and by reading some books.

This story started very early with discussions on fractals and chaos I had with some close friends but a turning point was some talks organised in Grenoble, France, where I did my engineering and doctoral studies (2000–2006). Particularly I remember two talks by Jacques Demongeot about dynamical systems in biology, that were very inspiring, and an excellent talk by Yves Colin de Verdière on ergodicity. In parallel, I started reading some books by Benoit Mandelbrot about fractals [8] and I had a very nice discussion with Jacques Viret (TIMC Laboratory, Grenoble), who was starting to write his book about topology and psychology [13]. He even had the kindness to give me a first draft of this book for revision. Jacques Demongeot, Jacques Viret and other mathematicians from TIMC were directly inspired by some works by René Thom. I also had discussions on dynamical systems, theoretical biology and complexity with other colleagues, notably Loic Forest and Nicolas Glade.

Later on, in 2008–2009, when I was doing a postdoc in Paris, under the supervision of Miguel Fernández, I worked on time-parallel time-integration for coupled partial differential equations (it was motivated by fluid-structure interactions in blood flows). The methods we studied underwent stability problems for conservative systems, like elastodynamics, and more generally for hamiltonian systems with symplectic geometric structure. This allowed me to discover many issues about numerical ordinary differential equations, the works of Martin Gander, Yvon Maday and other mathematicians, and the wonderful books by Ernst Hairer and colleagues. After two years of postdoc, I found no solution for this problem (I am still active on this issue in my free hours, and I still try some ideas with little success).

Around 2016, Yves Renard and I were invited to publish a paper [5] for a special issue of the journal *Advanced Modeling and Simulation in Engineering Sciences*, on the topic *Towards new paradigms for time integrators in computational dynamics*. This allowed me to study in more depth the Störmer–Verlet scheme and how it can be applied to contact in elastodynamics. I also had the opportunity to discuss with Laurent Monasse (at the WCCM 2016 in New York) and Frédéric Marazzato (I invited him for a talk in Dijon later on) about the new explicit time-marching scheme with pseudo-hamiltonian structure that they designed with Christian Mariotti and Alexandre Ern [9].

Around the year 2020, I was invited by Kanty Rabenorosoa to co-supervise Emerson Takematsu, a master student from Brazil, on modelling and simulation for soft robotics. It involved mathematical models of thin structures (rods) and bifurcation analysis. I had some interactions with Florence Bertails and Sébastien Neukirch who are outstanding contributors to the topic, see for instance their joint work [2], and the work of Sébastien Neukirch with Basile Audoly, about spaghettis [1] (Ig Nobel Prize 2006). They suggested I have a look at the book by R. Seydel to start understanding the topic of numerical bifurcations.

I had also some nice discussions with Christian Bonatti in the coffee room of the IMB (Dijon), particularly about the Brouwer fixed point theorem, and I met a lot of Latin American

students he worked with. Before leaving Dijon, I had the first opportunity to give lectures there on numerical ordinary differential equations, to students of the international master of mathematical physics [4].

Now, and since 2024, in Montevideo, I have the chance to hear some talks by colleagues from the dynamical systems group, particularly by Sofía Llavayol, Roberto Markarian, Alessandro Passeggi and Raphael Potrie, and also to have some discussions with Françoise Dal'Bo, Alessandro Passeggi, Sofía Llavayol and Mauricio Velasco (see for instance [6] and [10] for the general context). Most of the discussions are about the practical detection and characterization of chaos, due to recent breakthrough from Alessandro and his colleagues [3, 11]. This inspired a research project (PREXI Pedeciba) that I carried out with Guillermo Bustos, a talented undergraduate student who was under my supervision in 2025. On the other hand, I had the chance to start other related activities, such as lectures on numerical differential equations (with a related book that has been published by the Dirac editions), discussions with Marcelo Forets, Jorge Perez and Juan Pablo Borthagaray about reachability analysis and discussions with Benjamin Georgette and Cédric Pozzolini about Poisson's structures for continuum thermomechanics.

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2 Basic references

The two main references of the seminar are the following.

2.1 Hairer, Lubich and Wanner

The monograph of Hairer, Lubich and Wanner [7], first published in 2002 is by now a reference on geometric numerical integration. The book develops the modern theory of structure-preserving time integrators for ordinary differential equations: rather than viewing a numerical scheme merely as an approximation tool with a given order of accuracy, the authors promote the viewpoint that a good integrator should inherit the geometric structure of the flow it approximates. This leads to a systematic study of symplectic methods for Hamiltonian systems.

2.2 Seydel

A possible entry point to the numerical side of bifurcation analysis is the book of Seydel [12], whose third edition summarises three decades of development of the field. The central object of the book is the computation of stationary solutions and their stability, and the systematic numerical detection of the bifurcations through which solution branches lose or exchange stability: turning points, branch points, Hopf bifurcations and the routes they open towards periodic orbits and chaotic regimes. The distinctive feature of the approach is that bifurcation analysis is formulated as a numerical continuation problem: solution branches are traced by combining Newton-type solvers with predictor–corrector path-following strategies, and bifurcation points are detected and located as special solutions of augmented systems.

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3 Some key notions

The seminar is structured around the following topics.

3.1 Time integration for Hamiltonian dynamics

Many physical systems – planetary motion, elasticity, wave phenomena – are described by a function $H(q,p)$ of positions q and momenta p, called the *Hamiltonian*, and evolve according to Hamilton's equations

$$\dot{q} = \frac{\partial H}{\partial p}(q,p), \qquad \dot{p} = -\frac{\partial H}{\partial q}(q,p). \tag{1}$$

This seemingly simple structure has a remarkable consequence: the flow of (1) preserves the total energy $H(q(t),p(t))$ and the volume in phase space. A natural question is whether a numerical scheme can inherit these properties. The answer is generally no for classical schemes: for instance, the explicit Euler method applied to a harmonic oscillator ($H = \frac{1}{2}(p^2+q^2)$) produces a discrete orbit spiralling outwards, so the computed energy grows without bound. In contrast, the *symplectic Euler* method, which updates p before q using the fresh value,

$$p_{n+1} = p_n - h\,\frac{\partial H}{\partial q}(q_n, p_{n+1}), \qquad q_{n+1} = q_n + h\,\frac{\partial H}{\partial p}(q_n, p_{n+1}), \tag{2}$$

preserves the symplectic structure of the flow and, by backward error analysis, behaves over exponentially long times like the exact flow of a slightly perturbed Hamiltonian. This is the central theme of [7] and the guiding principle of structure-preserving integration: *the discretization should respect the geometry of the equation.*

3.2 Numerical integration on Lie groups

Some systems are constrained to live on a curved set carrying a group structure – rotations of a rigid body, unit quaternions for orientation, orthogonal matrices – collectively called Lie groups. A numerical method that treats the coordinates as if they lived in a flat space $\mathbb{R}^d$ produces iterates that leave the group, destroying the invariants of the problem. The remedy is to express the motion *through* the group using the exponential map. If G is a matrix group and $\mathfrak{g} = T_eG$ its tangent space at the identity, the flow of a vector field on G can be advanced by solving an equation in the tangent space and mapping the result back:

$$y(t) = \exp\big(\xi(t)\big)\, y(0), \qquad \xi(t) \in \mathfrak{g}, \tag{3}$$

where exp is the matrix exponential. This is the principle of *Lie group methods*: the update of ξ is computed by a classical scheme, while the exponential guarantees that $y(t)$ remains on G at every step.

3.3 Reachability analysis

The last decade has seen the rise of a complementary question: rather than computing or simulating individual trajectories, one asks what is the *set of all possible states* that a system can attain. This is the subject of *reachability analysis*. The motivation is twofold. First, many systems are subject to uncertainty: uncertain initial conditions, uncertain parameters, or inputs chosen by an external agent. Second, many applications – verification of safety properties, control synthesis, air-traffic management, robotics – require guarantees that hold for *all* trajectories at once, not merely for simulated ones. A typical safety question reads: *is it true that every trajectory starting in the initial set X_0 avoids the unsafe set U for all times $t \in [0,T]$?*

The mathematical formulation can be as follows. Given a discrete dynamics $x_{n+1} = \varphi(x_n)$ and a set of initial states X_0, the *reachable set* after N steps is defined recursively by

$$\mathcal{R}_0 = X_0, \qquad \mathcal{R}_{n+1} = \varphi(\mathcal{R}_n) := \{\varphi(x) : x \in \mathcal{R}_n\}, \tag{4}$$

and the safety question becomes $\mathcal{R}_n \cap U = \varnothing$ for all $n \le N$.

3.4 Numerical computation of bifurcations

When a differential equation depends on a parameter λ,

$$\dot{x} = f(x, \lambda), \qquad x \in \mathbb{R}^n,\ \lambda \in \mathbb{R}, \tag{5}$$

the structure of its equilibrium solutions – points x with $f(x, \lambda) = 0$ – can change qualitatively as λ crosses special values, called *bifurcation points*. The simplest example is the one-dimensional equation

$$\dot{x} = \lambda - x^2, \tag{6}$$

which has two equilibria $x = \pm\sqrt{\lambda}$ for $\lambda > 0$, one stable and one unstable, and no equilibrium at all for $\lambda < 0$: at $\lambda = 0$ a pair of equilibria is born (*saddle-node bifurcation*). Numerical bifurcation analysis aims to *compute* such phenomena: one traces the branch of equilibria with a continuation method (a predictor–corrector algorithm driven by Newton's method), monitors *test functions* whose vanishing signals a bifurcation, and solves augmented systems to locate the bifurcation point precisely and switch to the emerging branch. This is the algorithmic core of [12]. The difficulty, and the interest of the subject, is that the bifurcation is a global property of the branch: no local information at a single point reveals it.

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